

Assignment -1

MTH 314-Multivariate Analysis 2024

- ✓ 1. Draw random samples of sizes 5, 50, 200, 500, and 1000 from $N_3(\underline{\mu}, \underline{\Sigma})$

where $\underline{\mu} = \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix}$, $\underline{\Sigma} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 3 & 2 \\ 1 & 2 & 2 \end{pmatrix}$

and (i) establish $E(\bar{X}) = \underline{\mu}$ and $E(S) = \underline{\Sigma}$. Use any s/w language to do this.

2. Partition the random vector $\underline{X} = (X_1, X_2, X_3, X_4, X_5)'$ of order 5×1 into two components $\underline{X} = (X_1, X_2)'$ such that $\underline{X}_1 = (X_1, X_4)'$ and $\underline{X}_2 = (X_2, X_3, X_5)'$. The random vector $\underline{X}$ is having a multivariate normal distribution $N_5(\underline{\mu}, \underline{\Sigma})$ with mean vector and covariance matrix as follows:

$$\underline{\mu} = \begin{pmatrix} 2 \\ 15 \\ 6 \\ 20 \\ -10 \end{pmatrix}, \quad \underline{\Sigma} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 & 0 \\ 0 & 0 & 10 & 0 & 0 \\ 0 & 0 & 0 & 20 & 0 \\ 0 & 0 & 0 & 0 & 30 \end{pmatrix}$$

- ✓(i) Write the suitable partitioned vectors $\underline{\mu}_1, \underline{\mu}_2$ and matrices $\underline{\Sigma}_{11}, \underline{\Sigma}_{12}, \underline{\Sigma}_{21}$ and $\underline{\Sigma}_{22}$
(ii) Compute $E(\underline{X}_1 | \underline{X}_2 = \underline{x}_2)$, $E(\underline{X}_2 | \underline{X}_1 = \underline{x}_1)$, $V(\underline{X}_1 | \underline{X}_2 = \underline{x}_2)$, and $V(\underline{X}_2 | \underline{X}_1 = \underline{x}_1)$
(iii) Simulate $\underline{X}_1 | \underline{X}_2 = \underline{x}_2$, and $\underline{X}_2 | \underline{X}_1 = \underline{x}_1$ by draw random samples of sizes 5, 50, 200, 500, and 1000 from $N(\underline{\mu}, \underline{\Sigma})$ and establish the unbiased properties of the estimators of $E(\underline{X}_1 | \underline{X}_2 = \underline{x}_2)$, $E(\underline{X}_2 | \underline{X}_1 = \underline{x}_1)$, $V(\underline{X}_1 | \underline{X}_2 = \underline{x}_2)$, and $V(\underline{X}_2 | \underline{X}_1 = \underline{x}_1)$

- 3. Draw the contour diagram of $\underline{X} = (X_1, X_2)'$ from $N_2(\underline{\mu}, \underline{\Sigma})$ by generating the random numbers where $\underline{\mu} = \begin{pmatrix} 2 \\ 10 \end{pmatrix}$, $\underline{\Sigma} = \begin{pmatrix} 2 & 3 \\ 3 & 15 \end{pmatrix}$.

- ✓ 4. Let the random vector $\underline{X} = (X_1, X_2, X_3)'$ is having a multivariate normal distribution $N_3(\underline{\mu}, \underline{\Sigma})$. Find the distribution of $\begin{pmatrix} X_1 - X_2 \\ X_2 - X_3 \end{pmatrix}$.

- ✓ 5. If the random vector $\underline{X}$ is having a multivariate normal distribution $N_p(\underline{\mu}, \underline{\Sigma})$ with $|\underline{\Sigma}| \neq 0$, Show that the joint density can be written as the product of marginal densities of $\underline{X}_1$ and $\underline{X}_2$ if $\underline{\Sigma}_{12} = 0$ where $\underline{X}_1$ and $\underline{X}_2$ are of orders $q \times 1$ and $((p - q) \times 1)$ respectively, and $\underline{\Sigma}_{12}$ is of order $((p - q) \times q)$.

6. ✓ Suppose $X^{(1)}$ and $X^{(2)}$ of q and $(p - q)$ components, respectively have the density

$$\frac{1}{(2\pi)^{\frac{p}{2}}} \exp\left(-\frac{Q}{2}\right), \quad \text{where}$$

$$Q = (x^{(1)} - \mu^{(1)})' A_{11} (x^{(1)} - \mu^{(1)}) + (x^{(1)} - \mu^{(1)})' A_{12} (x^{(2)} - \mu^{(2)}) \\ + (x^{(2)} - \mu^{(2)})' A_{21} (x^{(1)} - \mu^{(1)}) + (x^{(2)} - \mu^{(2)})' A_{22} (x^{(2)} - \mu^{(2)}).$$

Show that Q can be written as $Q_1 + Q_2$, where

$$Q_1 = [(x^{(1)} - \mu^{(1)}) + A_{11}^{-1} A_{12} (x^{(2)} - \mu^{(2)})]' A_{11} [(x^{(1)} - \mu^{(1)}) + A_{11}^{-1} A_{12} (x^{(2)} - \mu^{(2)})],$$

$$Q_2 = (x^{(2)} - \mu^{(2)})' (A_{22} - A_{21} A_{11}^{-1} A_{12}) (x^{(2)} - \mu^{(2)}).$$

(i) ✓ Show that the marginal density of $X^{(2)}$ is $\frac{|A_{22} - A_{21} A_{11}^{-1} A_{12}|^{1/2}}{(2\pi)^{\frac{p-q}{2}}} \exp\left[-\frac{Q_2}{2}\right]$.

(ii) ✓ Show that the conditional density of $X^{(1)}$ given $x^{(2)}$ is $\frac{|A_{11}|^{1/2}}{(2\pi)^{\frac{q}{2}}} \exp\left[-\frac{Q_1}{2}\right]$.